

Design FMEA — ArchF Drone Battery

Document Info

Field	Value
Document Number	VBF-T8R1MIMO-DFMEA-01
Cell Model	ArchF (Chen2020 modified)
Date	2026-08-31
Scope	Qualitative failure mode analysis (virtual design phase)

FMEA Table

Failure Effect	S	Potential Cause	O	Current Design Control
Reduced drone range	8	Electrode thickness too thin / inactive mass too high	2	calc-energy verification, mass optimization
Power loss during high-demand maneuvers	9	Electrode transport resistance too high	2	DFN simulation at 5C, boosted electrolyte conductivity
Overheating, degradation, safety risk	7	Increased overpotential from boosted transport	4	4C charge thermal simulation
Plating formation, internal short	10	High charge rate, low temperature, thick electrodes	2	Plating simulation at 4C/45°C
Assembly difficulty, weight budget exceeded	5	Electrode area fixed by parameter set	3	Mass calculation verification
Reduced operational life	6	SEI growth, electrode degradation	5	Aging simulation (not run for this case)
Capacity retention may differ from simulated	7	Transport parameters are domain estimates, not measured	4	Three-model heterogeneous voting (protocol Step 2)

Key Findings

- **Highest RPN:** Cycle life (210) — aging study not performed; recommend future iteration
- **Second highest:** Electrolyte transport uncertainty (140) — estimates need experimental validation
- **Thermal safety** (112) — marginal but documented; active cooling recommended
- **All primary design objectives met** with documented trade-offs